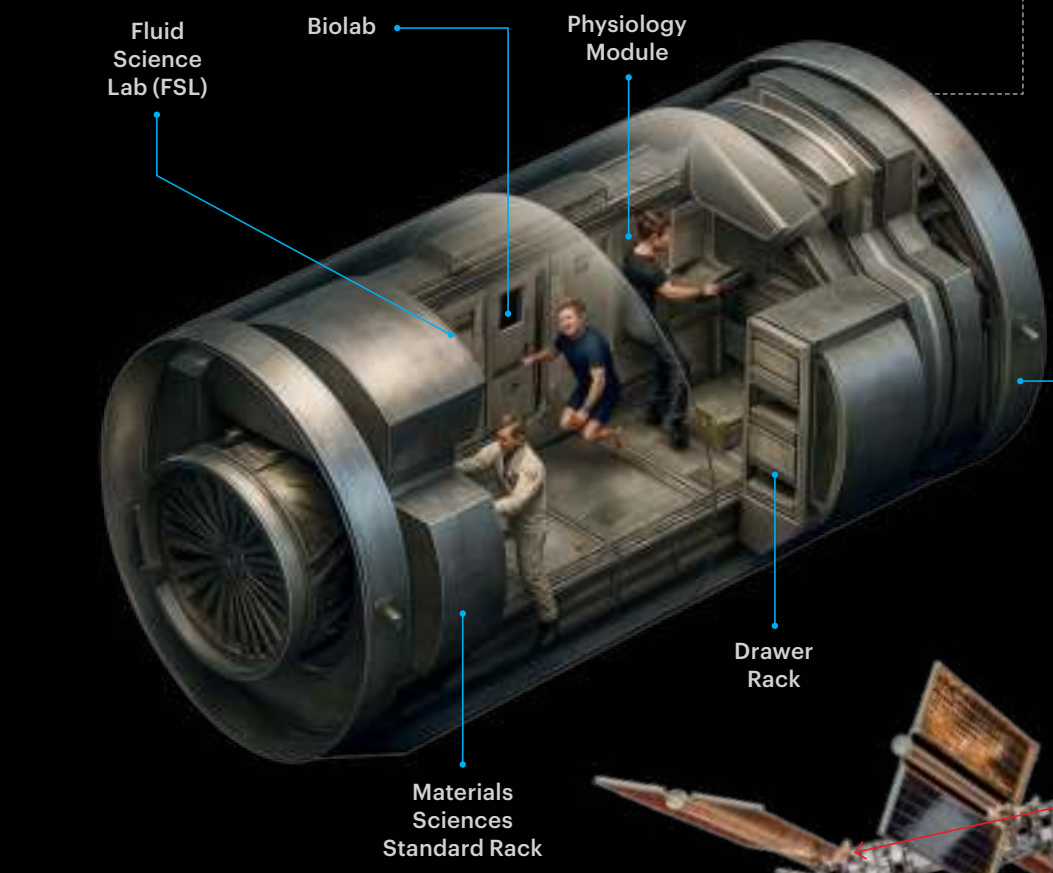
Wonders of an Orbiting Lab

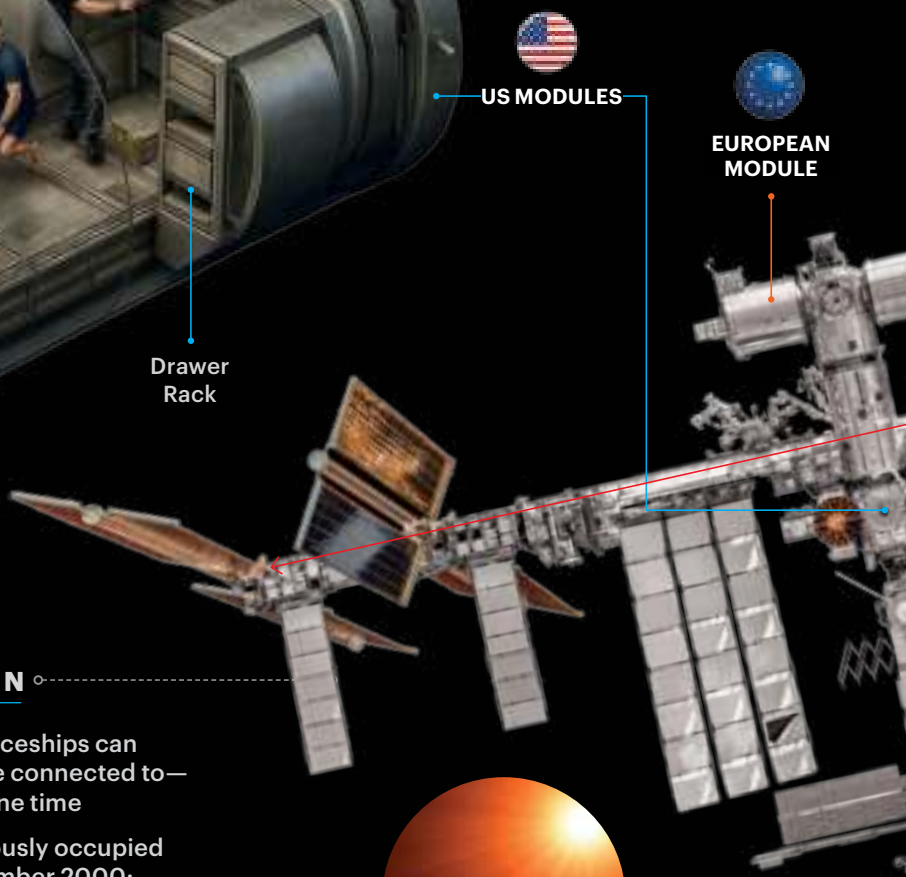
Indian astronaut Shubhanshu Shukla's two-week sojourn in the International Space Station is full of new learnings and experiments

Graphic by **NILANJAN DAS**



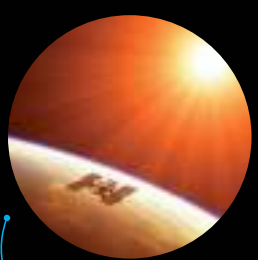
THE UNITY MODULE

- Second module in ISS, delivered to orbit by US space shuttle Endeavour in December 1998
- Provides living and working space; contains over 50,000 mechanical items, over 100 external and internal electrical cables
- It has six berthing ports, and has been connected to several other modules



INTERNATIONAL SPACE STATION

- The ISS is a permanently crewed orbital laboratory placed in low earth orbit (Avg altitude: 400 km)
- Orbits the Earth at a speed of 7.66 km/ sec, completing an orbit in 90 minutes or around 16 orbits a day
- The space station was set up between 1998 and 2011 after an initial US-Russia agreement in 1993
- It is a collaboration of 5 space agencies: NASA (US), Roscosmos (Russia), ESA (Europe), JAXA (Japan) and CSA (Canada)
- Eight spaceships can dock—or be connected to—the ISS at one time
- Continuously occupied since November 2000; counting Shukla et al, has hosted 280 astronauts from 23 nations
- ISS is composed of 16 pressurised modules: six Russian, eight American, one European and one Japanese
- Each module has diverse functions—scientific research, crew habitation, storage and spacecraft control



16

NUMBER OF SUNRISES AND SUNSETS VIEWED DAILY FROM THE ISS

With its velocity, it completes 16 orbits of the Earth in 24 hours